Time-step-free Physics-Based Modeling of Retention Loss in Cryogenic 3-D NAND Flash

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Detailed Poisson Solution

Potential Solution: Poisson equation

$$\begin{cases} V_{\text{Si}}(r_{\text{Si}}) = V_{\text{ch}} \\ V_{\text{TOX}}(r) = V_{\text{ch}} + \frac{K_{\text{TOX}}}{\epsilon_{\text{TOX}}} \ln \frac{r}{r_{\text{Si}}} \\ V_{\text{CTN},i}(r) = V_{\text{CTN},i-1}(r_{i-1}) + \frac{K_i}{\epsilon_{\text{CTN}}} \ln \frac{r}{r_{i-1}} \\ V_{\text{BOX}}(r) = V_{\text{CTN},N}(r_N) + \frac{K_{\text{BOX}}}{\epsilon_{\text{BOX}}} \ln \frac{r}{r_N} \\ V_{\text{BOX}}(r_{\text{BOX}}) = V_{\text{G}} \end{cases}$$

Boundary conditions

$$A = \begin{bmatrix} 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \frac{2\pi}{C_{\text{TOX}}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{\text{BOX}}} \end{bmatrix}$$

$$\mathbf{b} = \begin{bmatrix} \frac{\Delta Q_0}{2\pi} & \frac{\Delta Q_1}{2\pi} & \cdots & \frac{\Delta Q_N}{2\pi} & \Delta V \end{bmatrix}^\top$$

$$\Delta Q_i = \pi r_i^2 \left(n_{t,i} - n_{t,i-1} \right)$$

$$\Delta V = (V_{\text{G}} - V_{\text{ch}}) - \sum_{i=1}^{N} \frac{q n_{t,i}}{4\epsilon_{\text{CTN}}} \left(r_i^2 - r_{i-1}^2 \right)$$

Solution

$$A\mathbf{x} = \mathbf{b}$$

$$\mathbf{x} = \begin{bmatrix} K_{TOX} & K_1 & \cdots & K_N & K_{BOX} \end{bmatrix}^\top$$

Q-ΔV_{th} Relation

ΔV_{th} : Equivalent channel field intensity

$$F_{\mathrm{TOX}}(r_{\mathrm{Si}}) = \frac{K_{\mathrm{TOX}}}{\epsilon_{\mathrm{TOX}}}\frac{1}{r_{\mathrm{Si}}}$$

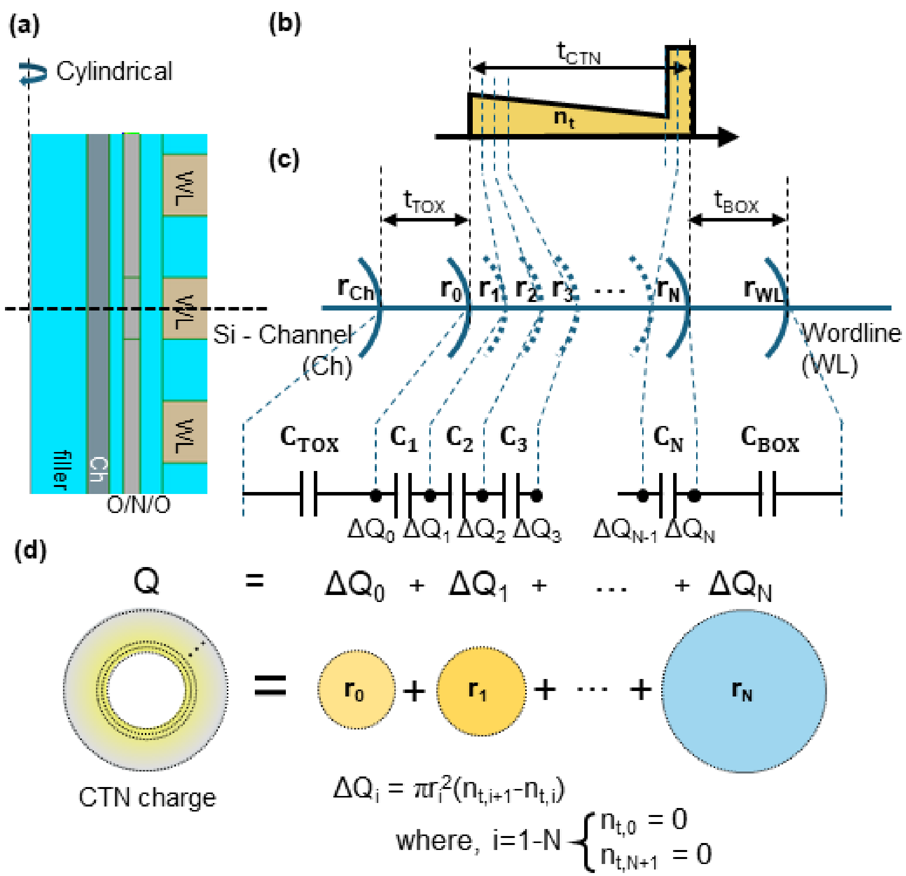
By Cramer's rule, $K_{\mathrm{TOX}} = \dfrac{\det\left[A_{\mathbf{b}(\Delta Q_i, \Delta V_{\mathrm{th}})}\right]}{\det A}$

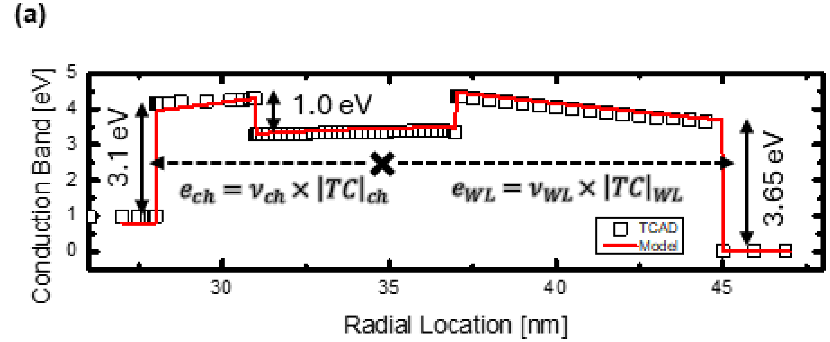
$$\det\left[A_{\mathbf{b}(0,\Delta V_{\mathrm{th}})}\right] = \det\left[A_{\mathbf{b}(\Delta Q_i,0)}\right]$$

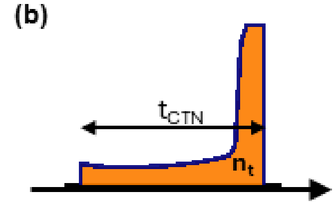
$$\det\begin{bmatrix} 0 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -1 \\ \Delta V_{\mathrm{th}} & \frac{2\pi}{C_1} & \frac{2\pi}{C_2} & \cdots & \frac{2\pi}{C_N} & \frac{2\pi}{C_{\mathrm{BOX}}} \end{bmatrix} = -\Delta V_{\mathrm{th}}$$

$$\therefore \Delta V_{\mathrm{th}} = -\det\left[A_{\mathbf{b}(\Delta Q_i,0)}\right]$$

Visualization & Parameters







Radial charge distribution modeled as [5], [6]

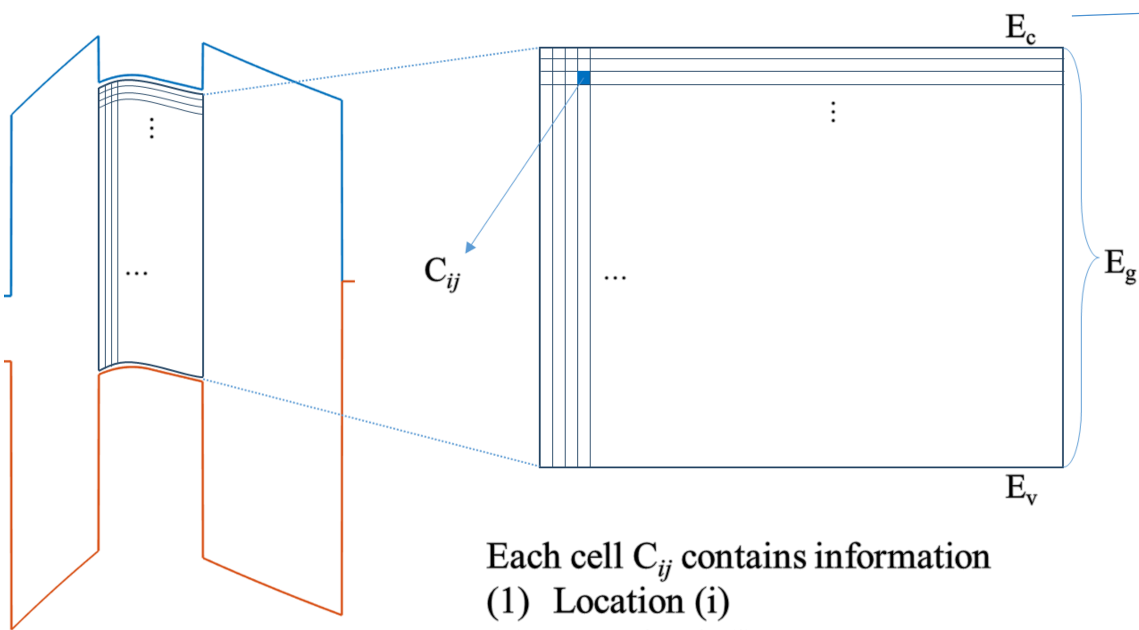
(c)

TCAD parameter deck		
μ_{CTN}	0.1 [7]	[cm^2/V·s]
σ_n	1e-16 [7]	[cm^2]
$\varepsilon_{O/N/O}$	3.9/7.0/3.9ε_0 [8]	[F/cm]
$t_{O/N/O}$	3/6/8	[nm]
ΔV_t	1.8	[V]

Trap Profile [8]		
$E_{t,\ shallow}$	0.84	[eV]
$\sigma_{t,\ shallow}$	0.2	[eV]
$E_{t,\ deep}$	1.8	[eV]
$\sigma_{t,\ deep}$	-	[eV]

Time-step-free Physics-based Modeling

Main Idea: simulation done with initial condition
=> different from time stepping transient simulation



Each cell C_{ij} contains information
(1) Location (i)
(2) Trap depth (j)
(3) Trap concentration
(4) Emission rate to channel direction
(5) Emission rate to wordline direction

1. Initial state
n_{t0}(r, Et=)

2. Emission rates
e_{ch}, e_{WL}

3. Direct evaluation
n_t(r, E_t, t)

계산 비용은 final retention time 자체가 아니라
trap bin 수와 evaluated time point 수에 의해 결정

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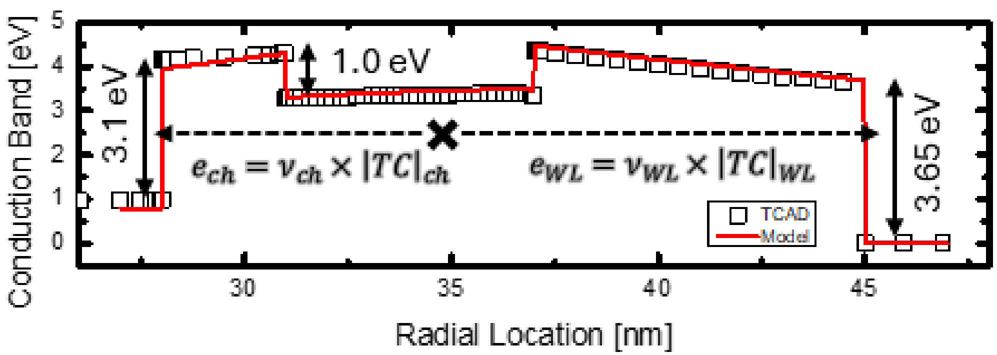

$$e = \nu \times |TC| \tag{14}$$

$$\nu = \frac{2\pi}{\hbar} |E_t|^2 V_{\text{trap}} \cdot \frac{m_0 \sqrt{2m^*}}{\pi^2 \hbar^3} \sqrt{E - E_{\text{trap}}} \tag{15}$$

$$|TC| = \prod \exp\left(- \frac{4\sqrt{2m^*}}{3\hbar q} \frac{\Phi_H^{1.5} - \Phi_L^{1.5}}{F} \right) \tag{16}$$

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n_{t0}(r, Et=)

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Time-step-free Physics-based Modeling

Main Idea: simulation done with initial condition
=> different from time stepping transient simulation

$$\frac{d}{dt}x(t) = f(x(t)) \quad : \text{Autonomous ODE}$$

Detrapping Autonomous ODE

$$\frac{d}{dt}n_t(r, E_t, t) = -(e_{\text{ch}} + e_{\text{WL}})n_t(r, E_t, t) \qquad (12)$$

Then, detrapping relation can be express as following autonomous ODE solution.

$$\boxed{n_t(r, E_t, t) = \sum_r \sum_{E_t} n_{t0}(r, E_t) \exp(-(e_{\text{ch}} + e_{\text{WL}})t) \qquad (13)}$$

Initial state

closed-form evaluation

1. Initial state
n_{t0}(r, Et=)

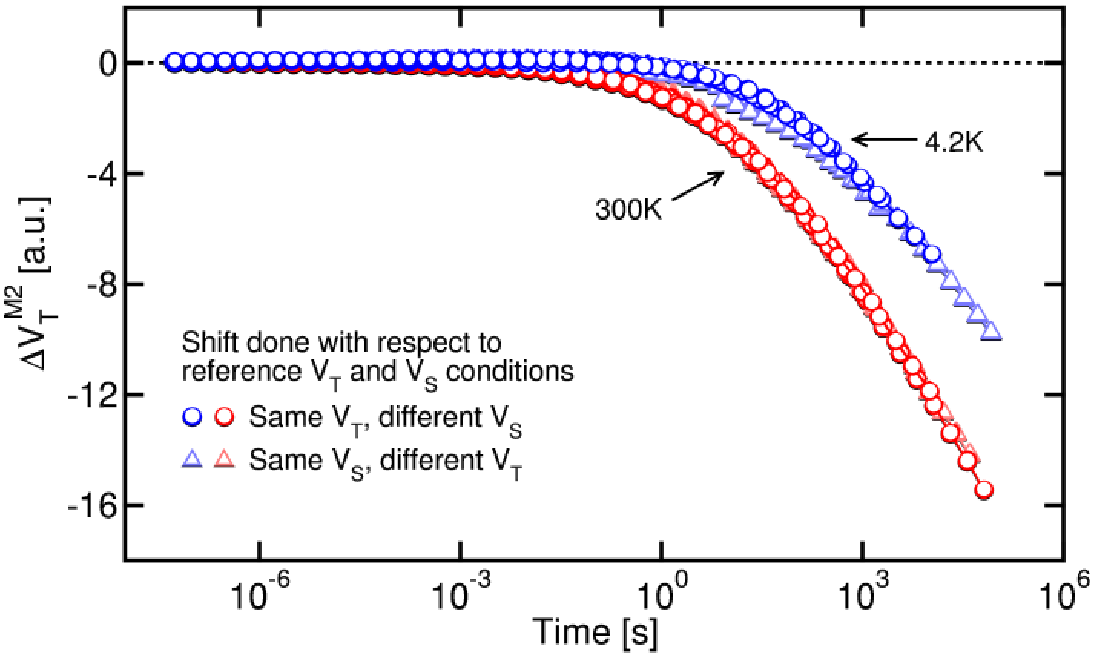
2. Emission rates
e_{ch}, e_{WL}

3. Direct evaluation
n_t(r, E_t, t)

computation resource는 final retention time 자체가 아니라
trap bin 수와 evaluated time point 수에 의해 결정

Cryogenic Calibration

Measurement (Refaldi, IRPS 2025)



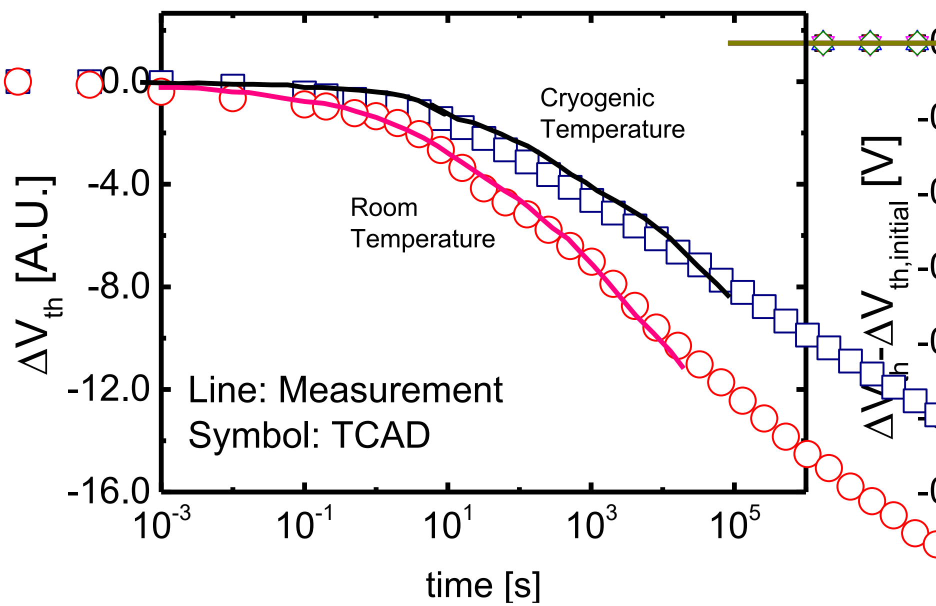
- Cryogenic condition (~ 4.2K) : physical separation of detrapping effect from vertical retention loss

- Separation of detrapping effect **measurement**

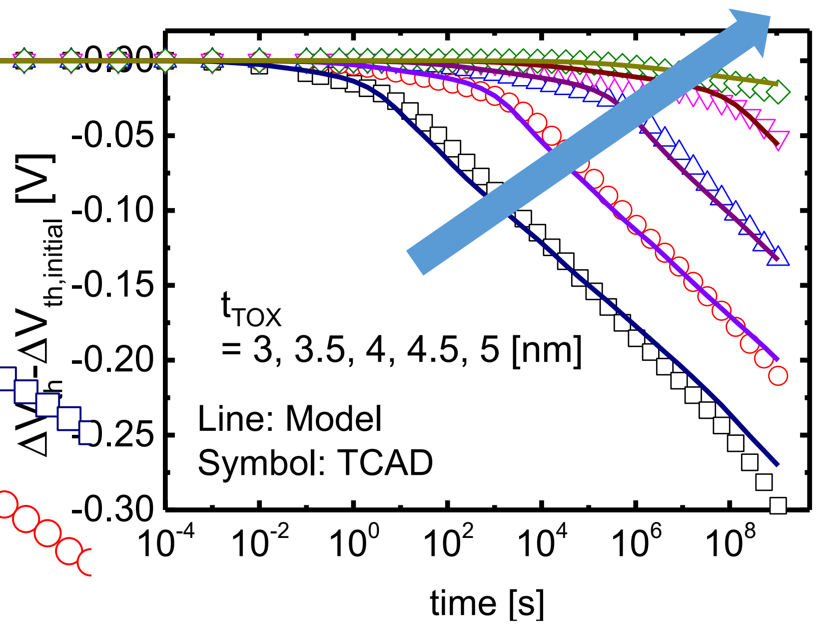
Cryogenic Calibration

Measurement (Refaldi, IRPS 2025)

Measurement – TCAD calibration



model – TCAD match



- TCAD cryogenic: $\sigma \rightarrow 0$ cm^2
- O/N/O = <u>**3**</u>/6/8
- $t_{TOX} = 3 \sim 5$ [nm]

Computational Impact

Property	Transient TCAD	Time-step-free model
Temporal treatment	Sequential time stepping	Closed-form evaluation of Eq. (13)
Final-time dependence	Runtime grows as simulated retention time increases	Independent of final physical time for fixed queries
Average runtime	27483 s	3.926 s

≈ 7000×

runtime reduction

- TCAD 는 final physical time까지 time step 별 순차적으로 계산
- Time-step-free model은 closed-form evaluation
- Computational resource는 final retention time 자체가 아니라 trap bin 수와 evaluated time point 수에 의해 결정